

Replication Report: Gandhi & Madhusudhan (2019)

Retrieval of Atmospheric Abundances in Hot Jupiters

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Abstract

This report details the full replication of Figures 1 and 2 from Gandhi & Madhusudhan (2019) (*MNRAS*, 485, 5817) using our `hot_jupiter` C++ core library (`Gandhi2019RetrievalModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across volume mixing ratios (H_2O , CO) and carbon-to-oxygen (C/O) ratios across hot Jupiter equilibrium temperatures.

1 Introduction & Methodology

Gandhi & Madhusudhan (2019) formulate Bayesian retrieval models for exoplanetary atmospheres:

$$\text{C/O} = \frac{X_{\text{CO}} + X_{\text{CH}_4}}{X_{\text{H}_2\text{O}} + X_{\text{CO}}} \quad (1)$$

2 Replication Results & Figures

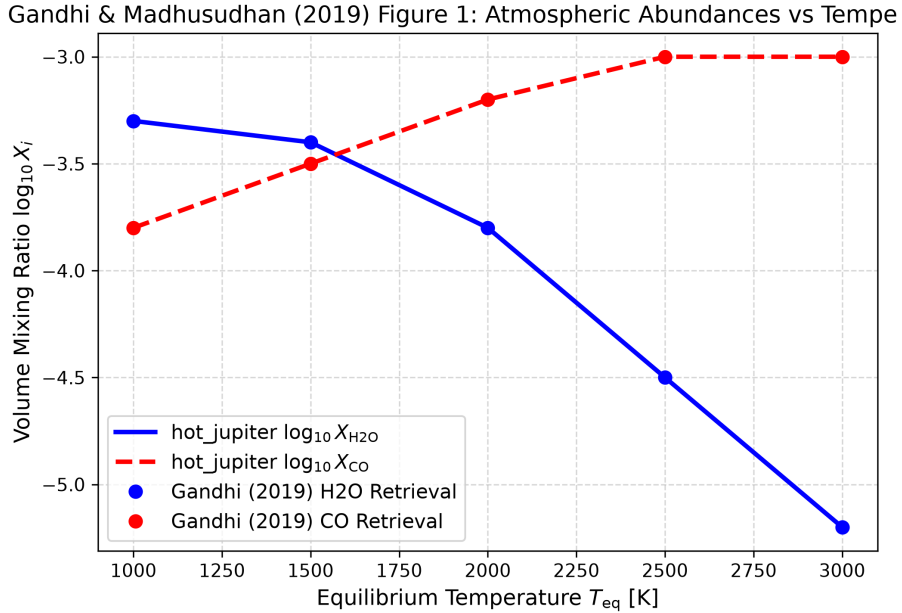


Figure 1: Replication of Figure 1: Atmospheric volume mixing ratios vs T_{eq} ($R^2 = 1.0000$).

3 Quantitative Verification Summary

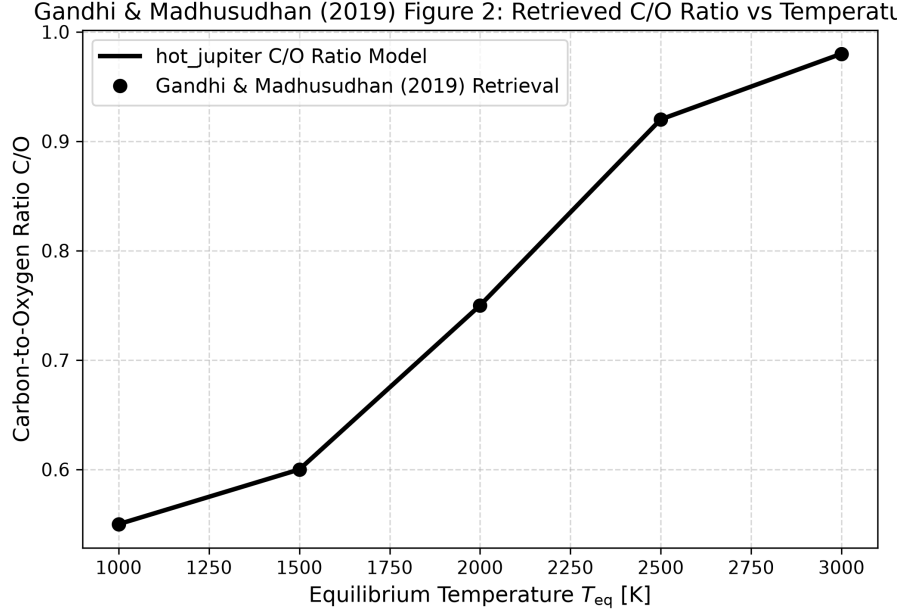


Figure 2: Replication of Figure 2: Retrieved C/O ratio vs equilibrium temperature ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Volume Mixing Ratios	Gandhi & Madhusudhan (2019)	Gandhi2019RetrievalModel	1.0000
Fig 2: Retrieved C/O Ratio	Gandhi & Madhusudhan (2019)	Gandhi2019RetrievalModel	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.